Taist Technical Platform Overview

A non-technical guide to the technology powering the Taist platform.

Platform Architecture

The Taist platform is built with modern, industry-standard technologies used by companies like Airbnb, Uber, and Shopify.

```
┌──────────────────────────────────────────────────────────────────┐
│                        TAIST PLATFORM                            │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│   ┌─────────────────┐        ┌─────────────────┐        ┌──────────┐  │
│   │   MOBILE APP    │        │     BACKEND     │        │ DATABASE │  │
│   │                 │◄──────►│     SERVER      │◄──────►│          │  │
│   │  iOS & Android  │        │                 │        │   MySQL  │  │
│   │  React Native   │  API   │  Laravel (PHP)  │        │          │  │
│   └─────────────────┘        └─────────────────┘        └──────────┘  │
│                                      │                                 │
│                                      ▼                                 │
│                          ┌──────────────────────┐                      │
│                          │   THIRD-PARTY SERVICES  │                   │
│                          │  Stripe • Firebase • AI │                   │
│                          └──────────────────────┘                      │
│                                                                        │
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```

Mobile Application

What It Is

A single codebase that runs on both **iPhone** and **Android** devices.

Technology

Component	Technology	What It Means
Framework	React Native + Expo	Write code once, deploy to both iOS and Android
Language	TypeScript	Modern JavaScript with built-in error checking
Navigation	Expo Router	Handles moving between screens in the app
State Management	Redux	Keeps app data consistent across all screens

App Store Presence

Platform	Store	Package ID
iOS	Apple App Store	`org.taist.taist`
Android	Google Play Store	`com.taist.app`

Current Version

- **App Version:** 29.0.0
- **iOS Build:** 10
- **Android Build:** 146

Backend Server

What It Is

The "brain" of the platform that processes all requests, handles business logic, and manages data.

Technology

Component	Technology	What It Means
Framework	Laravel 8	Popular, secure PHP framework used by millions of apps
Language	PHP 8.2	Server-side programming language
API Style	REST	Standard way for the app to communicate with the server
Authentication	OAuth 2.0 (Passport)	Secure login system with tokens

What The Backend Handles

- User registration and login
- Chef profiles and menu management
- Order processing and status updates
- Payment coordination with Stripe
- Notifications (push + SMS)
- AI-powered features
- Admin dashboard

Database

What It Is

Where all the platform's data is stored - users, orders, menus, reviews, etc.

Technology

Component	Technology	What It Means
Database	MySQL	Industry-standard relational database
Tables	19+ tables	Organized storage for different data types

Data Stored

```
┌──────────────────────────────────────────────────────────┐
│                      DATABASE CONTENTS                   │
├──────────────────────────────────────────────────────────┤
```

```
│                                                          │
│   USERS              ORDERS             MENUS            │
│   ─────              ──────             ─────            │
│   • Customers        • Order details    • Menu items     │
│   • Chefs            • Status tracking  • Descriptions   │
│   • Admins           • Payment info     • Prices         │
│   • Profiles         • Discounts        • Customizations │
│                                                          │
│   REVIEWS            MESSAGES           AVAILABILITY     │
│   ───────            ────────           ────────────     │
│   • Ratings          • Conversations    • Weekly schedules │
│   • Comments         • Order chats      • Day overrides  │
│   • AI variants      • Read status      • Time slots     │
│                                                          │
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```

Cloud Hosting (Railway)

What It Is

The servers that run the backend are hosted on **Railway** - a modern cloud platform that simplifies deployment and scaling. Railway is trusted by thousands of companies for its ease of use and reliability.

Server Setup

```
┌──────────────────────────────────────────────────────────┐
│                  RAILWAY INFRASTRUCTURE                  │
├──────────────────────────────────────────────────────────┤
│                                                          │
│   PRODUCTION                       STAGING               │
│   ──────────                       ───────               │
│   • Live customer traffic          • Testing environment │
│   • Real payments processed        • Safe for experiments │
│   • Auto-scaling                   • Development builds  │
│   • Zero-downtime deploys          • Preview deployments │
│                                                          │
│   Backend API                      Staging API           │
│   + MySQL Database                 + MySQL Database      │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

Why Railway?

Feature	Description
Simple Deployments	Push code to GitHub, Railway deploys automatically
Auto-Scaling	Handles traffic spikes automatically
SSL/HTTPS	All data encrypted in transit (automatic)
Managed Database	MySQL hosted and backed up by Railway

Zero Config	No server management needed

Third-Party Services

These are external services the platform integrates with to provide specialized functionality.

Payment Processing - Stripe

```
┌──────────────────────────────────────────────────────────┐
│                          STRIPE                          │
├──────────────────────────────────────────────────────────┤
│                                                          │
│   CUSTOMER PAYMENTS              CHEF PAYOUTS            │
│   • Credit card processing       • Direct bank deposits  │
│   • Secure card storage          • Automatic transfers   │
│   • Fraud protection             • Earnings dashboard    │
│   • Refund handling              • Tax documentation     │
│                                                          │
│   PCI Compliant = Bank-level security                    │
│                                                          │
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```

Notifications - Firebase & Twilio

Service	Purpose
Firebase Cloud Messaging	Push notifications to phones
Twilio	SMS text messages

When Notifications Are Sent:

- New order placed (to chef)
- Order accepted/rejected (to customer)
- Order completed
- New message in chat
- Order reminders

AI Features - OpenAI

Feature	What It Does
Menu Descriptions	Auto-generates professional dish descriptions
Grammar Check	Fixes spelling and grammar in text
Review Generation	Creates varied reviews from authentic ones

Cost: Less than $1/month for typical usage

Maps & Location - Google Maps

Feature	What It Does

Location Search	Find chefs by ZIP code
Address Validation	Verify delivery addresses
Distance Calculation	Chef service area management

Development Environments

The platform runs in multiple environments to ensure quality and safety.

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┌──────────────────────────────────────────────────────────────────┐
│                    DEVELOPMENT PIPELINE                          │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│   LOCAL              STAGING              PRODUCTION             │
│   ─────              ───────              ──────────             │
│                                                                  │
│   Developer's        Test server          Live server            │
│   computer           for QA               for customers          │
│        │                  │                   │                  │
│        ▼                  ▼                   ▼                  │
│    ┌─────────┐         ┌─────────┐          ┌─────────┐         │
│    │  Code   │────────►│  Test   │─────────►│  Live   │         │
│    │ Changes │         │ Changes │          │ Release │         │
│    └─────────┘         └─────────┘          └─────────┘         │
│                                                                  │
│   Safe to break      Real data copy       Real customers         │
│   No real data       Test payments        Real payments          │
│                                                                  │
└──────────────────────────────────────────────────────────────────┘
```

Environment	Purpose	Data
Local	Developer testing on their computer	Fake test data
Staging	Quality assurance testing	Copy of real data
Production	Live app for real users	Real customer data

Version Control (GitHub)

What It Is

GitHub stores all the code and tracks every change ever made. Think of it as "Google Docs for code" with complete history.

How It Works

```
┌──────────────────────────────────────────────────────────────────┐
│                              GITHUB                              │
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│                                                                  │
```

```
│   CODE REPOSITORY                                          │
│   ───────────────                                          │
│   • All frontend code (mobile app)                         │
│   • All backend code (server)                              │
│   • Documentation                                          │
│   • Configuration files                                    │
│                                                            │
│   BENEFITS                                                 │
│   ────────                                                 │
│   • Complete history of all changes                        │
│   • Multiple developers can work simultaneously            │
│   • Rollback to any previous version if needed             │
│   • Code review before changes go live                     │
│   • Automatic backups                                      │
│                                                            │
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```

Repository Structure

```
taist-mono/
├── frontend/          ← Mobile app code (React Native)
├── backend/           ← Server code (Laravel PHP)
├── docs/              ← Documentation
└── scripts/           ← Automation tools
```

Technology Summary

Frontend (Mobile App)

Technology	Version	Purpose
React Native	0.81	Cross-platform mobile framework
Expo	54.0	Development tools and services
TypeScript	5.9	Type-safe JavaScript
Redux	5.0	State management

Backend (Server)

Technology	Version	Purpose
Laravel	8.x	PHP web framework
PHP	8.2	Server language
MySQL	8.0	Database
Apache	2.4	Web server

Infrastructure

Service	Provider	Purpose
Hosting	Railway	Server & database hosting
Payments	Stripe	Payment processing
Push Notifications	Firebase	Mobile notifications
SMS	Twilio	Text messages
AI	OpenAI	Smart features
Maps	Google	Location services

Security Overview

Layer	Protection
Transport	HTTPS/SSL encryption for all data
Authentication	OAuth 2.0 tokens, secure password hashing
Payments	PCI-compliant via Stripe (no card data stored)
Infrastructure	Railway's built-in security, automatic SSL
Code	Private GitHub repository, code reviews

Scalability

The platform is built to grow:

Aspect	Current	Can Scale To
Users	Hundreds	Thousands+
Orders/Day	Dozens	Hundreds+
Server	Railway	Auto-scales as needed

Scaling Benefits with Railway:

- Automatic scaling based on traffic
- No manual server management
- Easy to add more resources
- Built-in load balancing

Maintenance & Updates

Task	Frequency	Description
App Updates	As needed	New features, bug fixes

Server Updates	Monthly	Security patches
Database Backups	Daily	Automated snapshots
Monitoring	24/7	Automated alerts for issues

Summary

Taist is built on **proven, enterprise-grade technologies**:

- **Mobile:** React Native provides native iOS & Android apps from one codebase
- **Backend:** Laravel/PHP handles all business logic securely
- **Database:** MySQL stores all data reliably
- **Hosting:** Railway provides scalable, secure cloud infrastructure
- **Payments:** Stripe ensures PCI-compliant payment processing
- **AI:** OpenAI powers smart features at minimal cost

The platform follows industry best practices with separate staging/production environments, version control via GitHub, and automated security through Cloudflare.

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